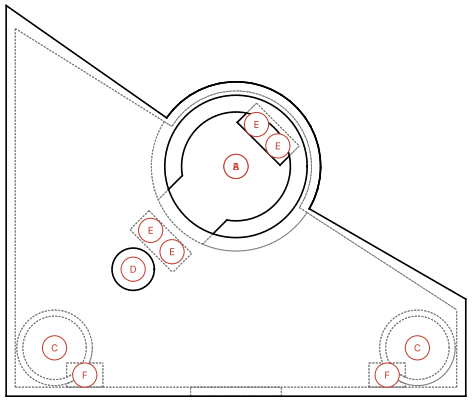
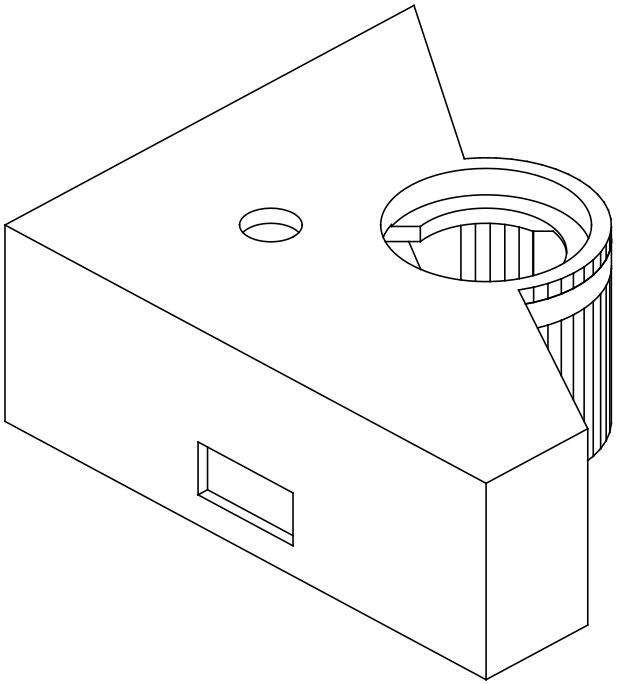
CA-101 Base

NOTES

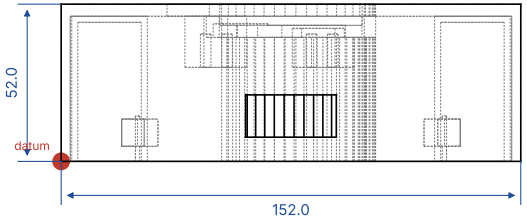
1. Hole sizes are the printed sizes from the PETG tolerance table (backend/cad/model.py TOL). Print CA-900 first and adjust TOL if a fit is off.
2. Screws thread straight into the printed pilots (HF kit #67675, no nuts).
3. Datum = the part's bounding-box corner (min X, Y, Z), lower left of the front and top views. Hole centres: (X, Y) for Z holes, (X, Z) for Y holes, (Y, Z) for X holes. ×k = k holes stacked along the axis; +n = n more positions (see the STEP file).
4. Shown in its assembly orientation; the STEP file defines all other geometry.

TOP VIEW

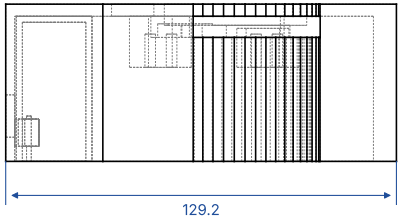




ISOMETRIC (not to scale)



FRONT VIEW



RIGHT VIEW

HOLE TABLE (letters mark hole centres on the view along that axis)

	Ø (mm)	×	axis	depth	centres from datum (mm)	fit / feature
A	47.10	1	Z	7.0	(76.0, 76.0)	6807 outer race press fit
B	36.00	1	Z	4.0	(76.0, 76.0)	bearing seat floor bore
C	21.00	2	Z	46.0	(16.0, 16.0); (136.0, 16.0)	ballast washer pocket
D	14.00	1	Z	4.0	(42.0, 42.0)	cable hole from the turret
E	3.50	4	Z	11.0	(47.8, 54.9); (54.9, 47.8) +2	#8-32 pilot
F	2.90	2	Z	15.0	(26.0, 7.0); (126.0, 7.0)	#6-32 pilot (self-threading)

PART NO.	**CA-101**
TITLE	**Base**
MATERIAL	PETG printed, 0.4 mm nozzle
MASS	109.2 g (CAD, 60 % fill)
QTY	1 per robot
SCALE	2:5
UNITS	mm · general tolerance ±0.2
SOURCE	cad/step/parts/base.step (from backend/cad/model.py)
PROJECTION	Third angle ⊕◁
SHEET	2 of 13
DRAWN	ME 2331 Project 2 team 2026-11-02